Representation Theory Seminar Notes

Oscillators and Quantization

Fall 2026

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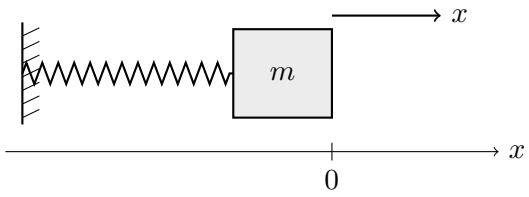
References **17**

1 The Harmonic Oscillator as the Simplest Nontrivial Integrable System

Speaker: Evgeny Mukhin
Date: September 11, 2026

1.1 The harmonic oscillator

Consider a mass $m > 0$ attached to a spring and moving along the real line. By Hooke's law and Newton's second law,



By convention, $x = 0$ is the equilibrium position. The constant $k > 0$ is the spring constant, and the minus sign in Hooke's law means that the force is restoring: it points toward equilibrium.

$$F = -kx, \qquad F = m\ddot{x}.$$

Therefore,

$$m\ddot{x} + kx = 0.$$

Writing

$$\omega = \sqrt{\frac{k}{m}} > 0,$$

the equation becomes

$$\ddot{x} + \omega^2 x = 0.$$

Its general real solution is

$$x(t) = A\cos(\omega t) + B\sin(\omega t) = \sqrt{A^2 + B^2}\,\sin(\omega t + \phi),$$

For $A^2 + B^2 > 0$, the phase is determined modulo 2π by

$$\sin\phi = \frac{A}{\sqrt{A^2 + B^2}}, \qquad \cos\phi = \frac{B}{\sqrt{A^2 + B^2}}.$$

When $A = B = 0$, the solution is identically zero and no phase is needed.

The total energy is

$$E = \frac{m}{2}\left(\dot{x}^2 + \omega^2 x^2\right) \geq 0.$$

Substitution of the solution gives

$$E = \frac{m\omega^2}{2}(A^2 + B^2),$$

so E is constant and $\dot{E} = 0$. This can also be checked directly from the equation of motion:

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + m\omega^2 x\dot{x} = m\dot{x}(\ddot{x} + \omega^2 x) = 0.$$

The point $A = B = 0$ is the equilibrium solution. Every other solution is periodic with period $2\pi/\omega$. For the classical oscillator and small oscillations, see [1, Chapter 5].

1.2 From energy to the Hamiltonian

The second-order equation for x can be rewritten as a first-order system by passing from configuration space to *phase space*. Normalize the mass to $m = 1$ and introduce coordinates

$$q = x, \qquad p = \dot{x}.$$

Thus q records position and p records momentum. In these variables, the energy becomes the function

$$H(p, q) = \frac{1}{2}\left(p^2 + \omega^2 q^2\right).$$

This function is the *Hamiltonian*. For this mechanical system, the Hamiltonian is exactly the total energy: the term $p^2/2$ is the kinetic energy, while $\omega^2 q^2/2$ is the potential energy. The original equation of motion is equivalent to Hamilton's first-order system

$$\dot{q} = p = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\omega^2 q = -\frac{\partial H}{\partial q}.$$

The number of *degrees of freedom* is the number of independent coordinates required to describe the configuration of the system. The mass–spring system moves along a line, so its configuration is determined by the single coordinate q and it has one degree of freedom. Its phase space nevertheless has dimension two, because one needs both q and its conjugate momentum p to specify a state. More generally, a system with configuration coordinates $q_1, \ldots, q_n$ has n degrees of freedom and a $2n$-dimensional phase space with coordinates

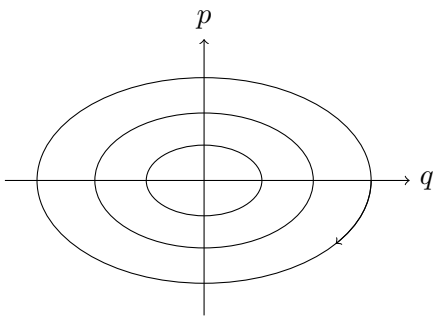
$$(p_1, \ldots, p_n, q_1, \ldots, q_n).$$

For example, a single unconstrained particle moving in three-dimensional space has three degrees of freedom and a six-dimensional phase space.

For a fixed positive energy value $E > 0$, the level set $H(p, q) = E$ is the ellipse

$$p^2 + \omega^2 q^2 = 2E, \qquad \frac{p^2}{2E} + \frac{q^2}{2E/\omega^2} = 1.$$

Here H denotes a function on phase space and E denotes its fixed value. The zero-energy level is the single equilibrium point.



Each ellipse is an orbit of the Hamiltonian flow. Increasing E enlarges the ellipse, while the equilibrium $(q, p) = (0, 0)$ is the zero-energy orbit. Since $\dot{q} = p$, the flow moves clockwise in the (q, p) convention used in the figure.

1.3 Observables and evolution

Let

$$\mathcal{O} = C^\infty(\mathbb{R}^2, \mathbb{R})$$

be the algebra of observables on phase space. Its elements are smooth functions $f(p, q)$, with addition and multiplication defined pointwise. In particular, $\mathcal{O}$ is a commutative, associative, unital algebra.

Before defining a bracket between two arbitrary observables, consider the operation determined by the Hamiltonian:

$$\{H, f\} := \frac{\partial H}{\partial p}\frac{\partial f}{\partial q} - \frac{\partial H}{\partial q}\frac{\partial f}{\partial p}.$$

For an observable f with no explicit time dependence, this operation describes its evolution. Along a solution $(p(t), q(t))$ of Hamilton's equations,

$$\begin{aligned}\frac{d}{dt} f(p(t), q(t)) &= \frac{\partial f}{\partial p}\dot{p} + \frac{\partial f}{\partial q}\dot{q} \\ &= -\frac{\partial f}{\partial p}\frac{\partial H}{\partial q} + \frac{\partial f}{\partial q}\frac{\partial H}{\partial p} = \{H, f\}.\end{aligned}$$

We write $\dot{f}$ for this derivative along a trajectory. Thus the Hamiltonian is not only the energy observable. Through the operation $f \mapsto \{H, f\}$, it generates the time evolution of every observable. In particular,

$$\dot{q} = \{H, q\} = \frac{\partial H}{\partial p}, \qquad \dot{p} = \{H, p\} = -\frac{\partial H}{\partial q}.$$

Thus f is an integral of motion precisely when

$$\dot{f} = \{H, f\} = 0.$$

Taking $f = H$ gives

$$\dot{H} = \{H, H\} = 0,$$

as follows immediately from the displayed formula. After the general bracket is defined below, this identity will also follow from its antisymmetry. Energy conservation is therefore built into the Hamiltonian formulation when H has no explicit time dependence. If $f = f(t, p, q)$ also depends explicitly on time, the total derivative is instead

$$\frac{d}{dt} f(t, p(t), q(t)) = \frac{\partial f}{\partial t} + \{H, f\},$$

where the bracket differentiates only the phase-space variables. The conservation criterion is also discussed in [2, Theorem 18.9].

1.4 The Poisson algebra of observables

Now let the phase space be $\mathbb{R}^{2n}$, with coordinates $(p_i, q_i)_{i=1}^n$, and let $\mathcal{O} = C^\infty(\mathbb{R}^{2n}, \mathbb{R})$. The operation motivated above extends to two arbitrary observables $f, g \in \mathcal{O}$ by

$$\{f, g\} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} - \frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} \right).$$

The canonical coordinates satisfy

$$\{p_i, q_j\} = \delta_{ij}, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, q_j\} = 0.$$

More abstractly, in these notes a *Poisson algebra* is a commutative, associative, unital algebra $\mathcal{O}$ over a fixed field $\urcorner$ (here $\mathbb{R}$ or $\mathbb{C}$), equipped with a $\urcorner$-bilinear operation

$$\{\,\cdot\,,\,\cdot\,\} : \mathcal{O} \times \mathcal{O} \longrightarrow \mathcal{O}$$

such that:

1. The bracket makes $\mathcal{O}$ into a Lie algebra. Explicitly, it is bilinear, antisymmetric,

$$\{f, g\} = -\{g, f\},$$

and satisfies the Jacobi identity

$$\{f, \{g, h\}\} = \{\{f, g\}, h\} + \{g, \{f, h\}\}.$$

2. The Lie bracket is compatible with the commutative multiplication through the Leibniz rule

$$\{f, gh\} = \{f, g\}\, h + g\, \{f, h\}.$$

The operation on $\mathcal{O} \times \mathcal{O}$ is bilinear; its equivalent realization on $\mathcal{O} \otimes_{\urcorner} \mathcal{O}$ is linear. The tensor product is algebraic and is taken over the same field $\urcorner$. Indeed, a bilinear map

$$\mathcal{O} \times \mathcal{O} \longrightarrow \mathcal{O}, \qquad (f, g) \longmapsto \{f, g\},$$

determines, by the universal property of the tensor product, a unique linear map

$$\mathcal{O} \otimes_{\urcorner} \mathcal{O} \longrightarrow \mathcal{O}, \qquad f \otimes g \longmapsto \{f, g\},$$

and conversely. Thus $\mathcal{O} \otimes \mathcal{O}$ does not mean that a bracket is taken only on pure tensors; it is the linear space through which every bilinear operation on two observables factors. For comparison, ordinary multiplication is another bilinear operation and similarly defines the linear map

$$m\colon \mathcal{O} \otimes_{\urcorner} \mathcal{O} \longrightarrow \mathcal{O}, \qquad f \otimes g \longmapsto fg.$$

As a concrete example of the Poisson bracket for one degree of freedom, take $f = p^2$ and $g = q^2$. Then

$$\left\{p^2, q^2\right\} = 4pq.$$

For scalars a, b, bilinearity gives

$$\left\{ap^2, bq^2\right\} = ab\left\{p^2, q^2\right\} = 4abpq,$$

which is precisely what allows the bracket to descend from pairs (f, g) to tensors $f \otimes g$.

Equivalently, for every fixed $f \in \mathcal{O}$, the map $g \mapsto \{f, g\}$ is a derivation of the commutative algebra $\mathcal{O}$. Antisymmetry gives the corresponding Leibniz rule in the first argument. Hence $C^\infty(\mathbb{R}^{2n})$, with pointwise multiplication and the bracket above, is a Poisson algebra. The dynamics associated with $H \in \mathcal{O}$ is the derivation

$$D_H = \{H, \cdot\}.$$

Its action gives $\frac{d}{dt} f(p(t), q(t)) = (D_H f)(p(t), q(t))$ for a time-independent observable f.

1.5 Symplectic formulation

The Hamiltonian vector field associated with f is

$$X_f = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial p_i} \frac{\partial}{\partial q_i} - \frac{\partial f}{\partial q_i} \frac{\partial}{\partial p_i} \right).$$

A one-form has the local expression

$$\alpha = \sum_{i=1}^{n} (f_i \, dq_i + g_i \, dp_i),$$

and the natural pairing with a vector field

$$X = \sum_{i=1}^{n} \left(\widetilde{f}_i \frac{\partial}{\partial q_i} + \widetilde{g}_i \frac{\partial}{\partial p_i} \right)$$

is

$$\langle X, \alpha \rangle = \sum_{i=1}^{n} (\widetilde{f}_i f_i + \widetilde{g}_i g_i).$$

Using the convention compatible with the Poisson bracket above, the standard symplectic form is

$$\Omega = \sum_{i=1}^{n} dq_i \wedge dp_i = -\sum_{i=1}^{n} dp_i \wedge dq_i.$$

It is closed and nondegenerate, and

$$\{f, g\} = -\Omega(X_f, X_g).$$

It is also exact. For example, if

$$\theta = -\sum_{i=1}^{n} p_i \, dq_i,$$

then $\Omega = d\theta$ and hence $d\Omega = 0$. Nondegeneracy means that if $\Omega(X, Y) = 0$ for every vector field Y, then $X = 0$. Consequently, a function f determines a unique Hamiltonian vector field. With the conventions above, it is characterized equivalently by

$$\iota_{X_f} \Omega = df.$$

The Poisson bracket can then be recovered from the symplectic form, so the Poisson and symplectic descriptions encode the same dynamics. For the geometric formulation, see [1, Chapters 8–9] and [2, Lecture 18].

1.6 Integrability

In general, Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

An autonomous Hamiltonian system on a real $2n$-dimensional symplectic phase space is *Liouville integrable* if it admits n smooth first integrals $H_1, \dots, H_n$ which are functionally independent and pairwise in involution. Conservation and involution mean

$$\dot{H}_i = \{H, H_i\} = 0 \quad (1 \le i \le n), \qquad \{H_i, H_j\} = 0 \quad (1 \le i, j \le n).$$

The first equality describes the derivative along every Hamiltonian trajectory. Each H_i may vary across phase space; its value is fixed along a given trajectory. Functional independence means

$$dH_1 \wedge \cdots \wedge dH_n \neq 0$$

on an open dense subset of phase space. In the examples here we take $H_1 = H$, so the conservation equations also follow from pairwise involution.

To discuss a common level set, choose numerical values $c_1, \dots, c_n$ and write

$$F^{-1}(c) = \{(p, q) : H_i(p, q) = c_i \text{ for } 1 \le i \le n\}, \qquad F = (H_1, \dots, H_n).$$

This is the setting in which $H_i = c_i$ denotes fixed values. If a connected component Λ of this level set is compact and the differentials dH_i are independent at every point of Λ, the Liouville–Arnold theorem identifies Λ with an invariant n-torus. Action-angle coordinates exist in a neighborhood of that component, and the motion on each such torus is linear in its angle coordinates. The compactness and regularity conditions are part of this torus conclusion. See [1, Chapter 10] and [2, Section 18.4, Theorem 18.12].

For the one-dimensional harmonic oscillator, $n = 1$, so the Hamiltonian itself provides the required integral. The system is therefore the simplest nontrivial example of a Liouville-integrable system. Its regular common level sets are precisely the ellipses above, which are one-dimensional tori (circles). For a positive energy value E, the action integral is

$$I(E) = \frac{1}{2\pi} \oint_{H=E} p \, dq = \frac{E}{\omega}.$$

Thus the action as a phase-space function is $I(p, q) = H(p, q)/\omega$. On the positive-energy region its conjugate angle evolves at constant speed $\dot{\varphi} = \omega$. The contour integral is taken once around the positive-energy orbit in the direction of the Hamiltonian flow; the angle coordinate is undefined at the equilibrium point.

As an example of a two-degree-of-freedom system, consider the Hénon–Heiles Hamiltonian

$$H = \frac{1}{2}(p_1^2 + p_2^2) + \frac{1}{2}(q_1^2 + q_2^2) + q_1^2 q_2 - \frac{1}{3} q_2^3,$$

introduced in 1964. Unlike the harmonic oscillator, the generic Hénon–Heiles system is not integrable.

2 The Harmonic Oscillator as the Simplest Nontrivial Integrable System: Talk 2

Speaker: Evgeny Mukhin
Date: September 18, 2026

The vibrations of a molecular bond motivate a quantum oscillator. Solving its eigenvalue problem leads to ladder operators and the Heisenberg algebra. We then connect this operator algebra to the classical Poisson algebra through deformation quantization.

2.1 Starting point: the classical oscillator

The classical mass and spring give a commutative algebra of observables with an additional Poisson bracket. For the algebraic discussion we use the complex polynomial subalgebra

$$A = \mathbb{C}[p, q], \qquad \{f, g\} = \partial_p f\, \partial_q g - \partial_q f\, \partial_p g, \qquad \{p, q\} = 1.$$

Ordinary multiplication is commutative: $pq - qp = 0$. This is compatible with $\{p, q\} = 1$, since multiplication and the Poisson bracket are different operations.

In Talk 1 the mass-one energy was $H_{\mathrm{cl}} = (p^2 + \omega^2 q^2)/2$. For the quantum calculation it is convenient to use the rescaled quadratic expression

$$K = p^2 + \omega^2 q^2 = 2H_{\mathrm{cl}}.$$

Its positive level sets are the same ellipses. However, as a generator of Hamiltonian evolution, K gives twice the vector field of H_{cl}: multiplying a Hamiltonian by two rescales time. Below we write H for the quantum operator corresponding to K.

2.2 Physical motivation: vibrations of the hydrogen molecule

The two nuclei in H_2 can oscillate about an equilibrium separation. To isolate this vibrational motion, fix the electronic state and ignore rotation and center-of-mass motion. The electronic energy together with nuclear repulsion gives an effective potential $V(r)$ for the internuclear separation r. At a stable equilibrium r_0,

$$V'(r_0) = 0, \qquad V''(r_0) = k > 0.$$

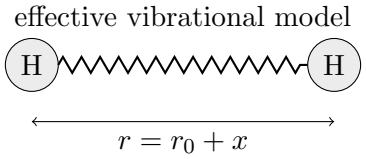
Writing $x = r - r_0$, Taylor expansion gives

$$V(r_0 + x) = V(r_0) + \frac{kx^2}{2} + O(x^3).$$

Thus, for small vibrations, a molecular bond behaves like a spring. With reduced mass $\mu = m_1 m_2/(m_1 + m_2)$ and $\omega = \sqrt{k/\mu}$, the vibrational energy relative to the minimum is approximately

$$H_{\mathrm{vib}} = \frac{p_x^2}{2\mu} + \frac{\mu\omega^2 x^2}{2}.$$

This is one vibrational degree of freedom, not a model of every motion of the whole molecule. In the harmonic approximation one extends the displacement coordinate to the full real line; this is a local approximation near the minimum, not a description of dissociation.



Quantum mechanics describes the vibrational state by a wavefunction in $L^2(\mathbb{R})$, rather than a single point of phase space. Its stationary energies are obtained from a differential-operator eigenvalue problem:

$$\left(-\frac{\hbar^2}{2\mu}\frac{d^2}{dx^2} + \frac{\mu\omega^2 x^2}{2}\right)\psi = E_{\text{vib}}\psi.$$

After mass scaling $q = \sqrt{\mu}\,x$ and $p = p_x/\sqrt{\mu}$, and doubling the energy operator, we obtain

$$H = \widehat{p}^{\,2} + \omega^2\widehat{q}^{\,2} = -\hbar^2\frac{d^2}{dq^2} + \omega^2 q^2.$$

An eigenvalue E of this doubled operator corresponds to physical vibrational energy $E_{\text{vib}} = E/2$. We now rename the mass-scaled coordinate q as x in the differential realization; this x is no longer the original unscaled internuclear displacement. The algebraic connection with the classical Poisson bracket will be made only after solving this quantum model. The physical background and transition to the quantum oscillator are discussed in [3, Lectures 7–9].

2.3 Position, momentum, and the eigenvalue problem

On wavefunctions set

$$(\widehat{q}f)(x) = xf(x), \qquad (\widehat{p}f)(x) = -i\hbar f'(x), \qquad \hbar > 0.$$

The commutator is $[S,T] = ST - TS$. Applied to a test function,

$$\begin{aligned}[\widehat{p},\widehat{q}]f &= -i\hbar(xf)' + i\hbar xf' \\ &= -i\hbar f,\end{aligned}$$

so

$$\boxed{[\widehat{p},\widehat{q}] = -i\hbar I.}$$

Here I is the identity operator. In operator formulas, a scalar denotes that scalar times I. From now on hats on p, q may be suppressed in operator formulas.

We use an inner product linear in the first argument:

$$(f,g) = \int_{\mathbb{R}} f(x)\overline{g(x)}\,dx.$$

On rapidly decreasing smooth functions (the Schwartz space $\mathcal{S}(\mathbb{R})$), integration by parts gives

$$(xf,g) = (f,xg), \qquad (\partial_x f, g) = -(f,\partial_x g).$$

Thus the formal adjoints satisfy $q^* = q$, $\partial_x^* = -\partial_x$, and $p^* = p$. Consequently,

$$(Hf,f) = \hbar^2\|f'\|^2 + \omega^2\|xf\|^2 \geq 0.$$

These operators are unbounded, so they are not defined on all of L^2. The calculations here take place on the common invariant core $\mathcal{S}(\mathbb{R})$; the spectral statements refer to the self-adjoint closure of H. Formal symmetry by itself is not a proof of self-adjointness. For the operator formulation, see [4, Sections 8.1 and 8.3].

Our problem is to find the nonzero square-integrable solutions of

$$Hf = Ef.$$

This leads naturally to a factorization of H.

2.4 Factorization and ladder operators

Assume $\omega > 0$ and define

$$a = \sqrt{\omega}\, q + \frac{i}{\sqrt{\omega}} p = \sqrt{\omega}\, x + \frac{\hbar}{\sqrt{\omega}} \partial_x, \qquad a^* = \sqrt{\omega}\, q - \frac{i}{\sqrt{\omega}} p = \sqrt{\omega}\, x - \frac{\hbar}{\sqrt{\omega}} \partial_x.$$

Expanding the products without commuting p and q prematurely,

$$aa^* = \omega q^2 + \omega^{-1} p^2 + i(pq - qp) = \frac{H}{\omega} + \hbar I,$$
$$a^* a = \omega q^2 + \omega^{-1} p^2 - i(pq - qp) = \frac{H}{\omega} - \hbar I.$$

Therefore

$$\boxed{[a, a^*] = 2\hbar I, \qquad \widetilde{H} := a^* a = \frac{H}{\omega} - \hbar I.}$$

In particular,

$$H = \omega(a^* a + \hbar I), \qquad (Hf, f) = \omega \|af\|^2 + \hbar\omega \|f\|^2.$$

This strengthens the earlier lower bound to $H \geq \hbar\omega I$.

The identity $[AB, C] = A[B, C] + [A, C]B$ yields

$$[\widetilde{H}, a^*] = 2\hbar a^*, \qquad [\widetilde{H}, a] = -2\hbar a.$$

Thus a^* raises the eigenvalue of $\widetilde{H}$ by $2\hbar$, while a lowers it by $2\hbar$ whenever its result is nonzero. The ladder-operator method is developed in [3, Lecture 9] and [4, Section 8.3].

2.5 The vacuum and the Fock ladder

For $f \in \mathcal{S}(\mathbb{R})$, the identity $(\widetilde{H} f, f) = \|af\|^2$ shows that $\widetilde{H} f = 0$ if and only if $af = 0$. The vacuum found below belongs to this common invariant core. The equation $af = 0$ is

$$\hbar f'(x) + \omega x f(x) = 0.$$

Set $\beta = \omega/\hbar$. Its solutions are scalar multiples of $e^{-\beta x^2/2}$. Choose the normalized vacuum

$$|0\rangle = \psi_0(x) = \left(\frac{\beta}{\pi}\right)^{1/4} e^{-\beta x^2/2}.$$

It satisfies

$$a|0\rangle = 0, \qquad \widetilde{H}|0\rangle = 0, \qquad H|0\rangle = \hbar\omega|0\rangle.$$

First define the unnormalized ladder vectors by $v_n = (a^*)^n|0\rangle$. The commutator identity

$$[a, (a^*)^n] = 2n\hbar(a^*)^{n-1} \qquad (n \geq 1)$$

follows by expanding the commutator as a sum of n terms. It gives

$$av_n = 2n\hbar v_{n-1}, \qquad a^* v_n = v_{n+1}, \qquad \widetilde{H} v_n = 2n\hbar v_n.$$

The annihilation formula holds for $n \geq 1$, with $av_0 = 0$; the other two formulas hold for all $n \geq 0$. For $n \geq 1$ the norms satisfy $\|v_n\|^2 = 2n\hbar\|v_{n-1}\|^2$. Starting from $\|v_0\| = 1$ gives $\|v_n\|^2 = (2\hbar)^n n!$ for all $n \geq 0$, so none of these vectors is zero. The normalized states

$$|n\rangle = \frac{v_n}{\sqrt{(2\hbar)^n n!}}$$

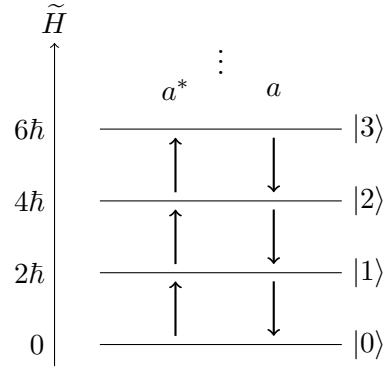
therefore obey

$$a|n\rangle = \sqrt{2n\hbar}\,|n-1\rangle \qquad (n \geq 1),$$
$$a^*|n\rangle = \sqrt{2(n+1)\hbar}\,|n+1\rangle \quad (n \geq 0).$$

Their energies are

$$\boxed{H|n\rangle = (2n+1)\hbar\omega|n\rangle, \qquad n = 0, 1, \ldots.}$$

For the physical Hamiltonian $H/2$, this is the familiar spectrum $E_n^{\text{phys}} = \hbar\omega(n + \frac{1}{2})$.



2.6 Hermite polynomials and completeness

Repeated creation produces a polynomial times the vacuum Gaussian. If $v_n = g_n(x)\psi_0(x)$, then

$$g_0 = 1, \qquad g_{n+1} = 2\sqrt{\omega}\,x g_n - \frac{\hbar}{\sqrt{\omega}} g_n'.$$

Indeed, differentiating the Gaussian contributes an additional $\sqrt{\omega}\,x g_n$ to $a^*(g_n\psi_0)$.

To fix the normalization, write $z = \sqrt{\beta}\,x$ and define the physicists' Hermite polynomials by

$$\mathsf{H}_n(z) = (-1)^n e^{z^2} \frac{d^n}{dz^n} e^{-z^2} = (2z - \partial_z)^n 1.$$

Then

$$g_n(x) = \hbar^{n/2}\mathsf{H}_n(\sqrt{\beta}\,x), \qquad \psi_n(x) = \frac{(\beta/\pi)^{1/4}}{\sqrt{2^n n!}}\,\mathsf{H}_n(\sqrt{\beta}\,x)e^{-\beta x^2/2}.$$

For example,

$$\mathsf{H}_0 = 1, \quad \mathsf{H}_1 = 2z, \quad \mathsf{H}_2 = 4z^2 - 2, \quad \mathsf{H}_3 = 8z^3 - 12z.$$

Their recurrence and derivative formulas are

$$\mathsf{H}_{n+1} = 2z\mathsf{H}_n - 2n\mathsf{H}_{n-1}, \qquad \mathsf{H}_n' = 2n\mathsf{H}_{n-1} \qquad (n \geq 1).$$

Two Hermite polynomial conventions. The probabilists' Hermite polynomials satisfy

$$\mathrm{He}_n(z) = (z - \partial_z)^n 1, \qquad \mathrm{He}_{n+1}(z) = z\mathrm{He}_n(z) - n\mathrm{He}_{n-1}(z) \quad (n \geq 1).$$

The initial polynomials are $\mathrm{He}_0 = 1$ and $\mathrm{He}_1 = z$. Their Rodrigues formula and relation to the physicists' polynomials are

$$\mathrm{He}_n(z) = (-1)^n e^{z^2/2} \partial_z^n e^{-z^2/2}, \qquad \mathsf{H}_n(z) = 2^{n/2} \mathrm{He}_n(\sqrt{2}\, z).$$

Both conventions are valid, but the variable and Gaussian weight must be rescaled together. See the two Hermite rows of DLMF, Table 18.9.1.

Orthogonality. Distinct eigenvalues of a self-adjoint operator give orthogonal eigenvectors. In polynomial language this is Gaussian-weighted orthogonality:

$$\int_{\mathbb{R}} \mathsf{H}_n(z) \mathsf{H}_m(z) e^{-z^2}\, dz = \sqrt{\pi}\, 2^n n!\, \delta_{nm}.$$

The polynomial weight is the *square* of the Gaussian in the wavefunction.

Completeness. The functions ψ_n form an orthonormal basis of $L^2(\mathbb{R})$. Here is a short completeness argument. If f is orthogonal to every ψ_n, the nonzero leading coefficients of the Hermite polynomials imply

$$\int_{\mathbb{R}} f(x) x^n e^{-\beta x^2/2}\, dx = 0 \quad \text{for all } n \geq 0.$$

The function $F(t) = \int_{\mathbb{R}} f(x) e^{-\beta x^2/2} e^{tx}\, dx$ is entire in t, by the Gaussian bound and Cauchy–Schwarz. All its derivatives at zero vanish, so $F = 0$. Restricting to imaginary t and using uniqueness of the Fourier transform gives $f(x)e^{-\beta x^2/2} = 0$ almost everywhere. Hence $f = 0$. Thus the ladder accounts for the full spectrum of the self-adjoint oscillator, not merely a collection of special eigenfunctions. See [4, Section 8.3] for the Hermite eigenfunctions and spectral completeness of the oscillator.

2.7 The Heisenberg algebra and its Weyl algebra representation

The relation $[a, a^*] = 2\hbar I$ has now emerged from the physical operator model and its factorization. It suggests studying the algebra generated by two symbols with precisely this relation.

More precisely, the Heisenberg *Lie algebra* has a basis u, v, z with

$$[u, v] = z, \qquad [u, z] = [v, z] = 0.$$

It acts on the oscillator states by $u \mapsto a$, $v \mapsto a^*$, $z \mapsto 2\hbar I$. The corresponding associative algebra with this fixed central character is

$$W_{2\hbar} = U(\mathfrak{h}_3)/(z - 2\hbar\, 1) \cong \mathbb{C}\langle u, v\rangle/(uv - vu - 2\hbar).$$

This is a Weyl algebra; for fixed $\hbar \neq 0$ a scalar rescaling identifies it with the standard first Weyl algebra $[\partial_y, y] = 1$. The Lie algebra and its enveloping-algebra quotient should not be conflated.

The algebraic Fock module is

$$\mathcal{F}_{\mathrm{alg}} = \mathrm{span}_{\mathbb{C}}\{v_n : n \geq 0\} \cong \mathbb{C}[y], \qquad v_n \longleftrightarrow y^n.$$

In this realization,

$$a^* = y, \qquad a = 2\hbar\frac{d}{dy}, \qquad |0\rangle = 1, \qquad \widetilde{H} = 2\hbar y\frac{d}{dy}.$$

For example, $[2\hbar\partial_y, y]f = 2\hbar f$. The positive coefficient is forced by our convention $[a, a^*] = 2\hbar I$. The scalar product transported from the oscillator is

$$(y^n, y^m) = \delta_{nm}(2\hbar)^n n!.$$

It makes multiplication by y adjoint to $2\hbar\partial_y$. Completing this algebraic module gives a Hilbert-space version of the Fock representation.

2.8 What it means to quantize a Poisson algebra

Only now do we return to the classical algebra and formulate the abstract connection. Let A be a unital commutative Poisson algebra over $\mathbb{C}$. A formal quantization is an associative $\mathbb{C}[[t]]$-algebra A_t with underlying module $A[[t]]$, whose product is of the form

$$f \star g = fg + tC_1(f, g) + t^2C_2(f, g) + \cdots, \qquad 1 \star f = f \star 1 = f,$$

and whose commutator satisfies

$$f \star g - g \star f = t\{f, g\} + O(t^2).$$

Here the C_j are bilinear maps $A \times A \to A$. In particular,

$$\boxed{\{f, g\} = C_1(f, g) - C_1(g, f).}$$

Thus $A_t/tA_t \cong A$, and the first-order failure of commutativity recovers the Poisson bracket. Equivalently, for lifts $\widetilde{f}, \widetilde{g}$,

$$\{f, g\} = \left.\frac{[\widetilde{f}, \widetilde{g}]}{t}\right|_{t=0}.$$

Changing a lift by a multiple of t does not change this reduction. The remainder must start at order t^2 to specify the first-order bracket.

The parameter t here is formal. We use a different letter from the physical positive number $\hbar$ to keep signs and specializations explicit. For special polynomial examples one can work over $\mathbb{C}[t]$ instead; general formal series need not be convergent or admit numerical specialization. Kontsevich's deformation quantization theorem addresses the general existence problem for Poisson manifolds. The examples below instead admit explicit algebraic constructions. For this quantization framework, see [6, Lecture 3].

2.9 A Lie algebra supplies a Poisson algebra

Let $\mathfrak{g}$ be a finite-dimensional complex Lie algebra with basis $e_1, \ldots, e_d$ and structure constants

$$[e_i, e_j]_{\mathfrak{g}} = \sum_k c_{ij}^k e_k.$$

Its symmetric algebra is the commutative polynomial algebra

$$S(\mathfrak{g}) = \mathbb{C}[e_1, \ldots, e_d],$$

equivalently the polynomial functions on $\mathfrak{g}^*$. Define a bracket first on generators by $\{e_i, e_j\} = [e_i, e_j]_{\mathfrak{g}}$ and extend by the Leibniz rule:

$$\boxed{\{f, g\} = \sum_{i,j,k} c_{ij}^k e_k \frac{\partial f}{\partial e_i} \frac{\partial g}{\partial e_j}.}$$

This is the linear Lie–Poisson (Kirillov–Kostant) bracket. Antisymmetry comes from the Lie bracket. Its Jacobiator is a derivation in each argument, so it suffices to check Jacobi on generators, where it is exactly the Jacobi identity in $\mathfrak{g}$.

For the Heisenberg Lie algebra above,

$$S(\mathfrak{h}_3) = \mathbb{C}[u, v, z], \qquad \{u, v\} = z, \qquad \{u, z\} = \{v, z\} = 0.$$

The central variable z is a Casimir, so the ideal $(z-1)$ is a Poisson ideal. The quotient has bracket $\{u, v\} = 1$, the classical canonical example.

2.10 Universal enveloping algebras and PBW

The quantization of this linear Poisson structure is

$$U_t(\mathfrak{g}) = T(\mathfrak{g})[[t]] \Big/ \overline{\langle xy - yx - t[x, y]_{\mathfrak{g}} : x, y \in \mathfrak{g} \rangle},$$

where $T(\mathfrak{g})$ is the tensor algebra and the bar means closure in the t-adic topology. Thus the Lie bracket is retained, but becomes the first-order commutator of a noncommutative multiplication.

The Poincare–Birkhoff–Witt (PBW) theorem says that ordered monomials

$$e_1^{\alpha_1} \cdots e_d^{\alpha_d}, \qquad \alpha_i \in \mathbb{Z}_{\geq 0},$$

give the corresponding basis over $\mathbb{C}[t]$ in the polynomial version, and a topological basis after completion. Hence

$$U_t(\mathfrak{g}) \cong S(\mathfrak{g})[[t]] \quad \text{as modules}, \qquad U_t(\mathfrak{g})/tU_t(\mathfrak{g}) \cong S(\mathfrak{g}).$$

PBW is essential: reducing the defining relations at $t = 0$ alone would not exclude hidden torsion or additional relations. Reordering adjacent generators proves spanning; independence is the substantive part of PBW. For enveloping algebras and the PBW theorem, see [5, Lectures 12–13].

One convenient identification is symmetrization:

$$\mathrm{Sym}(x_1 \cdots x_r) = \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

Transporting the enveloping multiplication through this module isomorphism produces a star product. For instance,

$$e_i \star e_j = e_i e_j + \frac{t}{2}[e_i, e_j]_{\mathfrak{g}},$$

so its first-order antisymmetric part is exactly the Lie–Poisson bracket.

For the ordinary enveloping algebra $U(\mathfrak{g})$, the degree filtration gives the related statement

$$\mathrm{gr}\, U(\mathfrak{g}) \cong S(\mathfrak{g}).$$

The polynomial family U_t interpolates between $S(\mathfrak{g})$ at $t = 0$ and $U(\mathfrak{g})$ at $t = 1$. This specialization statement refers to the polynomial family, not arbitrary formal power series.

2.11 Closing the circle: the oscillator algebra is a quantization

Return to $A = \mathbb{C}[p, q]$ with $\{p, q\} = 1$. Consider the polynomial family

$$W_t = \mathbb{C}[t]\langle P, Q\rangle/(PQ - QP - t).$$

Its ordered monomials $Q^r P^s$ form a $\mathbb{C}[t]$-basis. It is also the central reduction

$$W_t \cong U_t^{\mathrm{pol}}(\mathfrak{h}_3)/(z - 1), \qquad u = P, \quad v = Q.$$

Consequently $W_t/(t) \cong \mathbb{C}[p, q]$ and $[P, Q]/t = 1$ recovers $\{p, q\} = 1$.

There is an especially explicit product on symbols. Identify $q^r p^s$ with $Q^r P^s$. Normal ordering gives

$$\boxed{f \star g = \sum_{k\geq 0} \frac{t^k}{k!} (\partial_p^k f)(\partial_q^k g).}$$

For polynomial inputs the sum is finite. It is associative because it is transported from the associative Weyl algebra. At first order,

$$f \star g - g \star f = t(\partial_p f\, \partial_q g - \partial_q f\, \partial_p g) + O(t^2).$$

As a concrete continuation of Talk 1's example,

$$p^2 \star q^2 = p^2 q^2 + 4tpq + 2t^2, \qquad q^2 \star p^2 = q^2 p^2.$$

Thus $[P^2, Q^2] = 4tQP + 2t^2 I$, and its first-order symbol is $t\{p^2, q^2\} = 4tpq$. The higher-order term illustrates why one should not demand exact equality of a commutator and the classical bracket for all observables.

To recover the physical representation obtained earlier, specialize the polynomial parameter to

$$\boxed{t = -i\hbar, \qquad Q = x, \qquad P = -i\hbar\partial_x.}$$

The factor $-i$ is required by our convention $\{p, q\} = 1$. Then $P^2 + \omega^2 Q^2$ is precisely the quantum Hamiltonian already solved. The invertible change of generators

$$a = \sqrt{\omega}\, Q + \frac{i}{\sqrt{\omega}} P, \qquad a^* = \sqrt{\omega}\, Q - \frac{i}{\sqrt{\omega}} P$$

has commutator $[a, a^*] = 2itI$, which becomes $2\hbar I$. Before this specialization, a^* denotes the displayed algebraic generator. In the physical representation it is the adjoint of a on the common core. This is the promised connection: the algebra discovered in solving the molecular oscillator is a Weyl quantization of the classical oscillator's polynomial Poisson algebra. The Weyl and enveloping algebra examples are treated in [6, Section 3.6, Example 3.12].

2.12 From classical integrability to a quantum spectral problem

For an autonomous classical Hamiltonian H_{cl} on a phase space with n degrees of freedom, the goal is to find functionally independent first integrals $H_1 = H_{\mathrm{cl}}, H_2, \ldots, H_n$ with $\{H_i, H_j\} = 0$. They are conserved along the flow of H_{cl}. The quantum problem asks for lifts $\widehat{H}_i$ in a quantized algebra satisfying

$$[\widehat{H}_i, \widehat{H}_j] = 0,$$

and a representation with an appropriate scalar product, so that one can study joint eigenvalues and eigenvectors. Poisson commutativity only forces the first-order commutator to vanish; exact commuting lifts can require higher-order corrections and are an additional problem.

For unbounded operators, self-adjointness and suitable spectral commutativity must also be addressed; algebraic commutation on a core alone is not enough for a general joint spectral theorem. In this one-degree-of-freedom example, the single oscillator Hamiltonian and its complete Hermite basis give an explicit solution. The Fock scalar product is the concrete instance here of a compatible form: the creation and annihilation operators are adjoint to one another.

References

[1] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Graduate Texts in Mathematics **60**, Springer, 1989. Publisher's page. Chapter 5 discusses oscillations; Chapters 8–10 cover symplectic and Hamiltonian mechanics, Poisson brackets, integrability, and action-angle variables.

[2] A. Cannas da Silva, *Lectures on Symplectic Geometry*, Lecture Notes in Mathematics **1764**, Springer, 2001. Author's PDF (revised January 2006). Lecture 18 treats Hamiltonian dynamics and Poisson brackets. Theorem 18.9 characterizes first integrals; Section 18.4 and Theorem 18.12 discuss complete integrability and the Arnold–Liouville theorem.

[3] R. W. Field, *Physical Chemistry*, MIT 5.61, Fall 2017, Lectures 7–9. MIT OpenCourseWare lecture notes. Classical and quantum harmonic oscillators, molecular vibrations, and creation and annihilation operators.

[4] G. Teschl, *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, Graduate Studies in Mathematics **99**, American Mathematical Society, 2009. Author's PDF. Sections 8.1 and 8.3 cover position and momentum and the harmonic oscillator, including ladder operators, Gaussian ground states, Hermite eigenfunctions, and completeness.

[5] P. Etingof, *Lie Groups and Lie Algebras I*, MIT 18.745, Fall 2020, Lectures 12–13. MIT OpenCourseWare lecture notes. Universal enveloping algebras, the PBW theorem, and the associated graded description by the symmetric algebra.

[6] P. Etingof, *Lectures on Calogero–Moser Systems*, 2006, arXiv:math/0606233. Author's PDF. Lecture 3 develops deformation quantization. Section 3.6, especially Example 3.12, treats the Weyl algebra and universal enveloping algebras as quantizations of polynomial Poisson algebras.